

Device-Anchored Irrigation Logs: An Adversarial Evaluation of Near-Zero-Cost MRV for Low-Emission Rice

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Abstract. Measurement, reporting and verification (MRV) is the bottleneck keeping emission reductions in low-emission rice from becoming tradable credits, and monitoring is about three quarters of that cost. Because every credit-eligible water-management action is already an *actuation command*, a farmer-written irrigation log looks like free evidence of the highest rank under Decision 4801/QD-BNNMT. We built such a layer, proved it cannot accuse a compliant farmer, and then attacked it. The attack is the contribution. Replaying a log through the FAO-56 water balance under public ERA5 forcing exposes claimed depths no soil can hold, but an adversary who keeps the total volume and every individual depth intact and merely *shifts commands in time* inflates creditable dry phases by 69% and escapes both the physics layer and the pump meter in 91% of cases. We measure this over 281 audited logs at three Mekong Delta stations across two climate years, with the field deliberately violating every modelling assumption: soil texture across the FAO-56 Table 2 range, staged crop coefficient, irrigation efficiency 0.60–0.85, field rainfall differing from the grid cell, bund seepage, and farmers who forget, round and record late. Under those conditions the published point-model detector falsely accused 26.7% of honest farmers. Matching the log against the gateway’s signed, time-stamped command record instead catches 116/116 genuinely forged logs and accuses none. The result is negative but carries a repair: physics-based attestation cannot stand alone, and the timestamp must come from a device, not the farmer.

Keywords: alternate wetting and drying; monitoring, reporting and verification; carbon credit; low-emission rice; digital twin.

1 Introduction

Viet Nam has committed to one million hectares of low-emission rice by 2030, and the agronomy is settled: raising the severity of intermittent drying cut methane emission by 41–71% against continuous flooding in irrigated delta rice with no grain-yield penalty [1], and water regime is among the dominant controls on seasonal methane flux across Asian rice fields [2]. What is not settled is measurement. Under Decision 4801/QD-BNNMT the highest-ranked evidence for a credit claim is a record of the irrigation and drainage actions actually taken [3], and under Verra’s VM0051 the same record must survive third-party verification [4]. Producing it costs money that a small-holder cooperative does not have: monitoring alone accounts for about 77% of MRV expenditure in comparable programmes [5], and the current procedure relies on periodic photographs of a

water-level tube.

The obvious repair is to notice that a modern irrigation command is already a digital event. If a gateway actuates the pump, it can also record the actuation, and that record is a Tier-1 activity datum in the sense of Decision 4801 at no marginal cost. This is the premise of the companion work [6], which uses an FAO-56 twin as a safety gate deciding whether a command may be transmitted at all over a lossy channel.

We took the next step and asked whether the same twin could *audit* what a farmer claims after the fact, with no sensor deployed: a log of claimed depths is replayed through the water balance under public ERA5 forcing, and a claim that the soil could not have absorbed is evidence of fabrication. A first version of this layer reported no false accusation on 60 honest logs and detection of 60/60 forged ones. That result was correct and worthless, because the trajectories labelled “real” were generated by the same model that then audited them, with the same default soil profile, the same noiseless forcing, irrigation as a one-hour impulse of exactly {20, 40} mm, and a farmer who records the right depth at the right hour every time. The detector was grading its own homework.

This paper reports what happens when that circularity is removed. We rebuild the field so that it violates every assumption of the twin, and we replace the naive fraud patterns with an adversary that has read the detector. Three findings follow.

First, the point-model detector is not merely optimistic, it is *unsafe*: under realistic soil texture, staged crop coefficient, irrigation efficiency below unity, and recording noise, it falsely accuses 26.7% of honest farmers, because an absolute pump-gap threshold treats a farmer who forgets one entry or rounds to the nearest 5 mm exactly like one who irrigated secretly.

Second, and more damaging to the premise, physics-based attestation has a hole that no amount of modelling closes. An adversary who preserves the total pumped volume and keeps every individual claimed depth below the infiltration bound can simply *shift commands in time*. Total volume unchanged leaves the pump meter blind; individual depths unchanged leave the soil-capacity filter blind; and because the water balance integrates, the resulting trajectory is perfectly self-consistent. What changes is the number of dry phases of at least 72 hours — precisely the quantity that VM0051 and Decision 4801 pay for. We measure an inflation of 69% under the reduced uncertainty set and 78% under the full one, escaping both defence tiers in 91% of cases.

Third, the repair is cheap and it is not physical. Because the gateway that actuates the pump also holds a signed, time-stamped command record, the claimed log can be matched against what the device actually did. This catches 116/116 genuinely forged logs while accusing none of 60 honest ones. It is the only defence in our evaluation that reaches both figures, and it costs nothing beyond the gateway that the companion work already deploys.

The contributions are as follows.

1. *An adversarial evaluation of physics-attested MRV*. We formalise a threat model in which the adversary may alter the total volume, the individual depths and the timestamps of a claimed irrigation log, and we identify which of the three each defence tier can observe (Sec. 3, Table 1).

2. *A time-shifting attack that defeats soil-water physics.* We prove that any detector depending only on the replayed water balance is blind to a volume-preserving, depth-bounded permutation of claimed times, and we measure the resulting credit inflation at 69–78% (Sec. 5, Table 7).
3. *A device-anchored defence with a measured guarantee.* Matching the claimed log against the gateway command record catches every genuinely forged log in the evaluation and produces no false accusation (Sec. 8, Table 8).
4. *A robust accusation rule under model uncertainty.* Replacing the point model by an intersection over an uncertainty set Θ of 30 soil, crop-coefficient and efficiency configurations restores the no-false-accusation property when the true field is unknown, at a measured cost in detection power that we report rather than hide (Sec. 6.2, Table 6).
5. *Three corrections to previously published numbers.* The admissible depth bound is a range [49.0, 86.0] mm across Θ rather than a single 61 mm; the third physical filter cannot activate on dry-season delta rice because clipped depletion peaks at 0.9987; and a proportional pump-gap threshold is required, the absolute one having a 26.7% false-accusation rate (Sec. 6.1, Sec. 7, Table 2, Table 3).

The remainder of the paper gives the system model and the three filters (Sec. 2, Sec. 4), the guarantees and where they break (Sec. 6), the attack and the defence (Sec. 5, Sec. 8), the cost model (Sec. 10), and the evaluation (Sec. 11, Sec. 12).

2 System Model

2.1 Field water balance

The root-zone water content w_t [mm] evolves hourly under the FAO-56 bucket model [7]:

$$w_{t+1} = \text{clip}_{\leq FC}(w_t + P_t - RO_t - ET_{c,t} + u_t), \quad (1)$$

with crop evapotranspiration and a water-stress coefficient

$$ET_{c,t} = K_{s,t} K_c ET_{0,t}, \quad K_{s,t} = \frac{FC - w_t}{(1 - p) TAW}, \quad (2)$$

where $ET_{0,t}$ is the hourly reference evapotranspiration from the Penman-Monteith equation, P_t is rainfall, RO_t is surface runoff from the SCS curve-number method evaluated at daily resolution and shared pro rata over the hours of a rain event, and u_t is the irrigation depth applied in hour t . Drainage above field capacity FC is removed by the clipping operator, so mass conservation holds exactly and is checked as an invariant. The root-zone depletion fraction is

$$d_t = \frac{FC - w_t}{FC - WP}, \quad \text{span} = FC - WP, \quad (3)$$

with wilting point WP . For the reference silty clay loam of the delta, $TAW = 70$ mm, $FC = 160$ mm, $WP = 90$ mm, so $\text{span} = 70$ mm.

2.2 Replay of a claimed log

Let a claimed log supply the irrigation term of (1). The *replay operator* $\mathcal{R}$ maps a log to the trajectory it implies, by iterating (1) under the observed forcing $(P_t, ET_{0,t})$:

$$\mathcal{R}(L) = \{(\hat{w}_t, \hat{d}_t)\}_{t=0}^T, \quad (4)$$

with $\hat{w}_0 = w_{\text{init}}$ and

$$u_t = \sum_{(t_i, u_i) \in L} u_i \mathbb{1}[t_i = t], \quad (5)$$

so that $\hat{w}_t$ is the root-zone water content the log *claims* the field had, and $\hat{d}_t$ its depletion fraction under (3). The hat marks quantities reconstructed from the log rather than measured. Replay is deterministic and uses only public reanalysis, so two auditors replaying the same log obtain the same trajectory and the verdict is reproducible. A superscript minus denotes the state immediately *before* the claim of that hour is applied, as in $\hat{w}_t^-$, which is what a filter must inspect: the water available to receive an event.

The operator is also what makes the scheme cheap. No sensor is queried; the only inputs are the log, the public weather series and a soil profile. Its weakness, developed in Sec. 5, is that it can only check whether a trajectory is *possible*, never whether it *happened*.

3 Threat Model

A claimed irrigation log is a finite map

$$L = \{(t_i, u_i)\}_{i=1}^{|L|}, \quad u_i > 0, t_i \in [0, T), \quad (6)$$

where u_i is a claimed depth in millimetres and t_i an hour index. Three independent quantities can be forged, and each defence tier observes a different subset of them:

$$\underbrace{\sum_i u_i}_{\text{total volume}}, \quad \underbrace{\{u_i\}}_{\text{individual depths}}, \quad \underbrace{\{t_i\}}_{\text{timestamps}}. \quad (7)$$

Table 1 lists the resulting fraud classes. The adversary is *rational*: it maximises creditable dry phases subject to not being flagged, and it knows the detector, because the detector is published. Publishing a threshold is publishing the constraint the adversary optimises against; this is the sense in which ε in (12) below is not a secret.

Let M be the volume actually pumped, measured at the cooperative station, and $U(L) = \sum_i u_i$ the claimed total. The tier-1 statistic is

$$G = M - U(L), \quad (8)$$

and the credit-relevant quantity counts *dry phases*. Write $\mathcal{A}(\cdot)$ for the set of maximal intervals $[a, b)$ of a replayed trajectory over which the depletion stays at or above a threshold d_{dry} with no claimed irrigation, and which last at least 72 h — the three-day minimum of Decision 4801 [3]:

$$\begin{aligned} \mathcal{A}(\{\hat{d}_t\}) = \{[a, b) : & \min_{t \in [a, b)} \hat{d}_t \geq d_{\text{dry}}, \\ & u_t = 0 \ \forall t \in [a, b), \\ & b - a \geq 72 \text{ h}\}. \end{aligned} \quad (9)$$

The credited count is then

$$n_{\text{dry}}(L) = |A(\mathcal{R}(L))|, \quad (10)$$

with $d_{\text{dry}} = 0.45$ used throughout the evaluation. Revenue is proportional to n_{dry} , not to $U(L)$. This distinction is what makes the attack of Sec. 5 profitable, and it is invisible in a formulation that treats total volume as the target.

4 Physics-Based Filters and Their Exact Blind Spots

Replaying L through the FAO-56 root-zone bucket [7] under public ERA5 forcing produces a trajectory $(\hat{w}_t, \hat{d}_t)$; the companion work [6] gives the balance in full. Three filters apply.

V_1 , *saturation*. A claim issued while the root zone is already at field capacity cannot be stored:

$$V_1(L) = \{i : u_i > 0 \wedge \hat{w}_{t_i}^- \geq FC - \varepsilon\}. \quad (11)$$

V_4 , *storage excess*. A claim larger than the deficit available at that hour must have run off or percolated, so it was not “applied” in the sense the standard requires:

$$V_4(L) = \{i : u_i > \underbrace{(FC - \hat{w}_{t_i}^-)}_{\text{deficit}} + \varepsilon\}. \quad (12)$$

V_5 , *infiltration bound*. Independent of state, a single event of duration τ_i cannot deliver more than the soil can accept:

$$u_i \leq k_{\text{inf}} \tau_i + p_{\text{max}}, \quad (13)$$

with $k_{\text{inf}} = 10 \text{ mm h}^{-1}$ and $p_{\text{max}} = 100 \text{ mm}$ taken at the loose end of the published range for puddled rice, so that a farmer who records two adjacent events in one hour is not accused. At $\tau_i = 1 \text{ h}$ the bound is 110 mm.

A fourth filter, on depletion below the wilting point, was included in the published version of this layer. It cannot activate: see Sec. 7.

Proposition 1 (Exact blindness of aggregate defences). *Let L and L' share the same multiset of depths and the same total, i.e. $\{u_i\} = \{u'_i\}$ as multisets. Then*

$$G(L) = G(L') \quad \text{and} \quad |V_5(L)| = |V_5(L')|. \quad (14)$$

Proof. $U(L) = \sum_i u_i$ is a symmetric function of the multiset, so it is invariant under any reassignment of depths to hours; M is a property of the field, not of the log; hence $G = M - U$ is unchanged. Likewise (13) is evaluated per event and depends only on (u_i, τ_i) , never on t_i or on $\hat{w}$, so the count of violations is a function of the multiset alone. $\square$

Proposition 1 is the whole difficulty in one line. The pump meter and the infiltration bound are the two defences that do not depend on the model being right, and both are *exactly* invariant under the transformation the adversary needs. What remains is V_1 and V_4 , which do depend on order through $\hat{w}_{t_i}^-$. They turn out not to help.

5 The Time-Shifting Attack

Proposition 2 (Credit gain and evasion are the same action). *Let a claim at hour t have available deficit $c_t = FC - \hat{w}_t^-$. Between consecutive claims the bucket loses $\Delta_t = \sum_s (ETc_s + DP_s - P_s^+) \geq 0$, so delaying a claim by δ hours increases its deficit by the drainage accrued in those hours:*

$$c_{t+\delta} \geq c_t + \min(\delta \cdot \bar{\Delta}, FC - \hat{w}_t^-). \quad (15)$$

Consequently the transformation that moves claims out of a dry interval and onto its boundary simultaneously

- (i) increases n_{dry} , because the interval is no longer interrupted; and
- (ii) increases c_t at every moved claim, which by (12) makes V_4 less likely to fire.

There is no trade-off for the adversary to negotiate: the gradient of profit and the gradient of undetectability point the same way.

Proof. (i) is definitional: A in (10) requires no claim inside the interval, so removing one extends or creates a qualifying phase. For (ii), delaying a claim to $t + \delta$ means the bucket continues to lose water for δ further hours with no inflow, so $\hat{w}_{t+\delta}^- \leq \hat{w}_t^-$ and $c_{t+\delta} \geq c_t$; the bound (15) follows because the loss per hour is at least the mean $\bar{\Delta}$ until field capacity is reached, after which c saturates at $FC - \hat{w}$. Since V_4 fires only when $u_i > c_{t_i} + \varepsilon$ and u_i is unchanged, a larger c_{t_i} can only reduce the violation set. $\square$

This is why the attack cannot be closed by better hydrology. Any detector whose verdict is a function of the replayed trajectory must integrate the same drainage that makes the delayed claim look legitimate. The empirical version of the attack is a hill-climb over integer hour offsets in $[-36, 36]$ that accepts a move only if the total is preserved bit-for-bit, every depth stays under (13), and n_{dry} strictly increases.

Remark 1. *The attack does not steal water. Total pumped volume is unchanged, so the farmer pays the same and the atmosphere sees the same methane. What is stolen is history: the credit is issued for a dry-phase pattern that did not occur. Under VM0051 the crediting unit is precisely that pattern, so the loss is real even though no physical resource moved.*

6 When May a Farmer Be Accused?

6.1 The admissible depth is a range, not a number

The published version of this layer stated a single bound, $u \leq 61$ mm, as the largest claim that never raises an alarm. That number is an artefact of one soil profile and one initial condition. With V_4 firing when $u_i > c_{t_i} + \varepsilon$ and $c_t = d_t \cdot \text{span}$ where $\text{span} = FC - WP$, the bound at a field whose initial depletion is w_0 and whose operator irrigates no later than d^* is

$$u_{\max} = \max(d^*, w_0) \cdot \text{span} + \varepsilon. \quad (16)$$

Table 1. The three forgeable quantities of Eq. (7) and which fraud class alters each. “true” means the quantity is preserved, so the corresponding defence is blind by Prop. 1.

Fraud variant	Total mm	Each depth	Timing	What it forges
Honest farmer (field conditions)	true	true	true	Reference: seven reality-gap axes active
Added irrigation events	FALSE	true	true	Inflates the record of management activity
Infeasible claimed depth	FALSE	FALSE	true	One third of entries set to the whole storage span
Omission of real commands	FALSE	true	true	Deletes a 96 h window to manufacture a dry phase
Claim consolidation, 24 h	true	varies	FALSE	Same water, fewer and larger events
Claim consolidation, 96 h	true	varies	FALSE	Same water, events snapped to a coarse grid
Time shifting (rational)	true	true	FALSE	Hill-climbs each hour to maximise creditable dry phases
Rational adversary	true	true	FALSE	Chooses the consolidation window that maximises dry phases

Two corrections follow, both of which we verified by scanning the replay operator rather than by algebra alone. First, (16) uses max, not a difference: an initially dry field ($w_0 = 0.90$) admits *larger* claims than a wet one, and the previously published expression $(d^* - w_0)\text{span} + \varepsilon$ yields 43.5 mm for $\text{span} = 70$ mm and $w_0 = 0.25$, which corresponds to no physical state at all, since the operative deficit at that hour is either 0.25span (initially) or $d^*\text{span}$ (once the field has dried). Second, (16) is a bound *per event*, not per season: the bucket drains continuously through deep percolation and evapotranspiration, so a season-long claim total is not bounded by $u_{\max}$ and an adversary issuing many sub-threshold claims can inflate the total without bound. Table 2 gives the range that must be published instead, $[49.0, 86.0]$ mm across Θ .

6.2 Robust accusation under model uncertainty

The auditor does not know the farmer's soil, the growth stage, or the efficiency of the delivery system. Let Θ be the uncertainty set and let $V_\theta(L)$ be the violation set obtained by replaying under configuration θ . The published detector evaluated a single θ_0 (the default profile) and treated every violation as guilt. We replace it by an intersection:

$$V_{\text{rob}}(L) = \bigcap_{\theta \in \Theta} V_\theta(L), \quad (17)$$

$$\Theta = \mathcal{S} \times \mathcal{K} \times \mathcal{E} \times \mathcal{W},$$

with $|\mathcal{S}| = 5$ soil textures, $|\mathcal{K}| = 2$ crop-coefficient regimes, $|\mathcal{E}| = 3$ irrigation efficiencies and $|\mathcal{W}| = 5$ initial states, so $|\Theta| = 150$ and each log costs 150 replays. The semantics are deliberately one-sided:

$$a(L) = \begin{cases} \text{accuse,} & V_{\text{rob}}(L) \neq \emptyset, \\ \text{query,} & \text{otherwise.} \end{cases} \quad (18)$$

Proposition 3 (No false accusation within Θ). *If the true field configuration θ^* belongs to Θ and L is compliant under θ^* , then $V_{\text{rob}}(L) = \emptyset$ and the farmer is not accused.*

Proof. $V_{\text{rob}}(L) \subseteq V_{\theta^*}(L) = \emptyset$. □

Proposition 3 is what makes the layer deployable: model error can reduce detection power but cannot manufacture an accusation. The price is real and must be stated. Two implementation details matter, and getting either wrong silently breaks the proposition. A world that produces *no* violation must contribute the *empty set* to the intersection; skipping non-firing worlds computes $\bigcap$ over firing worlds only, which is an existential quantifier wearing the clothes of a universal one. And $\mathcal{W}$ must span the full range in which a claim can be legitimate: with $\mathcal{W} = \{0, 0.25\}$ the largest admissible depth anywhere in Θ is 22.5 mm, so every claim above that is “guilty in all worlds” and the intersection becomes vacuous — it fires on 102 of 150 worlds while 48 innocent worlds exist. Extending $\mathcal{W}$ to $\{0, 0.25, 0.5, 0.75, 0.9\}$ restores the intended semantics.

6.3 A proportional pump-gap threshold

The absolute rule $G > \tau_0$ with $\tau_0 = 5$ mm assumes the log is a faithful copy of what was pumped. It is not. Honest farmers forget entries, round to the nearest 5 mm, and record a day late. Each contributes to G without any wrongdoing, so the threshold must scale with the number of recorded events:

$$\tau(n) = \delta_{\text{rel}} M + z\sqrt{n} \sigma_{\text{event}}, \quad (19)$$

$$\sigma_{\text{event}}^2 = p(1-p)\bar{u}^2 + \frac{r^2}{12} + q(1-q)(s\bar{u})^2,$$

with p the forgetting rate, r the rounding step, q the depth-error rate, s its amplitude, $\bar{u} = 30$ mm a typical event depth, $\delta_{\text{rel}} = 2\%$ and $z = 3$. The three variance terms are the forget, round and depth-error contributions; the absolute rule is the degenerate case $n \rightarrow 0$, $\sigma_{\text{event}} \rightarrow 0$.

Table 2. The admissible depth bound Eq. (16) across the FAO-56 Table 2 range. The previously published single value 61 mm is one row of this table.

Soil (FAO-56 Table 2)	Span (mm)	$u_{\max}$ at $w_0 = 0$	$u_{\max}$ at $w_0 = 0.90$
sandy loam	55.0	49.0	54.5
loam	75.0	65.0	72.5
silt loam	90.0	77.0	86.0
silty clay loam	70.0	61.0	68.0
clay	80.0	69.0	77.0
Range to be published	—	49.0	86.0

Table 3. Reachability of the wilting-point filter over 150 configurations with an empty log, i.e. the driest forcing available.

Station	Year	Max clipped depletion	V_2 fires	Interpretation
can tho	2024	0.998	0	never below wilting point
can tho	2025	0.988	0	never below wilting point
soc trang	2024	0.999	0	never below wilting point
soc trang	2025	0.992	0	never below wilting point
ca mau	2024	0.991	0	never below wilting point
ca mau	2025	0.980	0	never below wilting point

7 A Filter That Cannot Fire

The published layer included a third physical filter, flagging depletion below the wilting point as physically impossible. It was doubly inoperative. Its threshold was set at 1.02 while the reported indicator $\hat{d}_t = \min(1, \max(0, \cdot))$ is clipped to $[0, 1]$, so the condition was unsatisfiable by construction: 0 firings in 281 logs. We repaired it by evaluating the unclipped depletion, and then measured whether the repaired filter can activate at all. Across three stations, two climate years, five textures and five initial states — 150 configurations — with an empty log, i.e. the driest possible forcing, the unclipped depletion peaks at 0.9987 (Table 3). A dry-season delta rice field does not reach the wilting point. The filter is therefore a *safety* device that guarantees no absurd state is accepted; it is not evidence of detection, and the layer must be described as having two operative physical filters and one inert one.

8 Device-Anchored Attestation

8.1 The gateway record

Proposition 1 shows that volume and depth are insufficient, and Proposition 2 shows that order is exactly what the adversary manipulates. The defence must therefore observe timestamps — and the only party that can vouch for a timestamp is the device that acted on it.

Let the gateway hold a signed command record

$$\mathcal{G} = \{(t_j, u_j, \sigma_j)\}, \quad (20)$$

where σ_j is a signature over (t_j, u_j) produced by the controller at actuation time. The anchor test compares the claimed log against it:

$$\begin{aligned} \Delta_{\text{time}} &= \{t_i\} \Delta \{t_j\}, \\ \Delta_{\text{depth}} &= \max_t |u(t) - u_{\mathcal{G}}(t)|, \end{aligned} \quad (21)$$

where Δ is the symmetric difference and Δ_{depth} is taken over hours common to both records. The verdict is then

$$A(L, \mathcal{G}) = \begin{cases} \text{forge}, & \Delta_{\text{time}} \neq \emptyset \vee \Delta_{\text{depth}} > \epsilon, \\ \text{match}, & \text{otherwise}, \end{cases} \quad (22)$$

with $\epsilon = 10^{-6}$ mm, i.e. exact agreement up to floating-point noise. Naming the two discrepancies separately matters for what follows: time-shifting alters Δ_{time} only, depth inflation alters Δ_{depth} only, and omission and addition alter both, which is why a single comparison catches all four. Time-shifting, consolidation, addition, omission and depth inflation all alter either the hour set or a depth at a common hour, so all are caught.

Proposition 4 (Completeness of the anchor). *If a claimed log L differs from the gateway record $\mathcal{G}$ in the multiset of hours or in any depth at a shared hour, then $A(L, \mathcal{G})$ returns forge. If L and $\mathcal{G}$ agree on both, then L is a faithful transcript and no forgery of the three quantities in (7) has occurred.*

Proof. The first clause is the definition of (22). For the second, total volume, individual depths and timestamps are all functions of the hour set and the per-hour depths, so agreement on those two fixes all three. $\square$

8.2 Why this does not re-introduce false accusation

Proposition 4 is strong enough to look dangerous: any discrepancy is treated as forgery, including the forgetting and rounding that Sec. 6 showed to be ubiquitous. The resolution is that the anchor does not compare a log against a *model*; it compares a transcript against the *source*. An honest farmer who forgets an entry is not accused of fraud, because the system of record is the gateway, not the farmer’s memory: the claim is simply completed from $\mathcal{G}$. In operational terms the anchor turns the log from *evidence the farmer must get right* into a *view of a record the device already holds*. False accusation then requires the gateway to disagree with itself, which is a device-integrity

question rather than a behavioural one, and is why the guarantee in Table 6 holds at 0/60 while the behavioural tiers hold at 26.7% and 5.0%.

8.3 Cost and the scope of the claim

The anchor requires the gateway of the companion work [6], which actuates the pump and already writes a signed, time-stamped command record as part of its safety function. The marginal cost of attestation is therefore the storage and transmission of a record that exists regardless, which is why the attested-log row of Table 4 remains near zero.

The scope must nonetheless be stated precisely, because it is narrower than the premise we started from. A log written by hand on a handset, with no device in the loop, is *not* admissible evidence on its own: Sec. 12 measures a rational adversary inflating credits by 69% while escaping every physics-based and volume-based test. The honest formulation of the result is that near-zero-cost MRV is achievable, but only where a controller stands between the farmer and the pump. Where irrigation is manual — a sluice gate opened by hand — the anchor is unavailable and this layer does not apply.

9 Subsidence Invariance

AWD compliance is usually stated as an absolute water-table or ponding depth, for example 15 cm below the surface, the reflooding threshold of the Safe-AWD treatment in [1]. In a subsiding delta the surface elevation $z_s(t)$ falls at rate $\dot{z}$, so an absolute threshold measured against a fixed benchmark drifts.

Proposition 5 (Drift of an absolute threshold). *Over a crediting period of T_c years an absolute threshold expressed against a fixed datum shifts by $\dot{z}T_c$. For $\dot{z} \in [1.1, 2.5] \text{ cm/yr}$ and $T_c = 5 \text{ yr}$ the shift is 5.5–12.5 cm, that is 37–83% of a 15 cm threshold. A criterion written in the depletion fraction (3) has zero drift, because FC and WP are soil properties referenced to the root zone and not to an absolute elevation.*

Proof. The absolute criterion compares a measured depth to a constant benchmark; if the benchmark and the surface move apart at $\dot{z}$, the effective threshold changes by $\dot{z}T_c$ over T_c years. The depletion criterion compares $FC - w_t$ to $(FC - WP)$, both of which are defined within the root zone and translate rigidly with the surface, so their ratio is invariant to z_s . $\square$

The subsidence rates are those measured for the delta by [8]. The practical consequence is that a scheme anchored on depletion fraction needs no re-surveying of field benchmarks during a five-year crediting period, whereas an absolute-depth scheme does.

10 Cost Model and Break-Even

Let a cooperative serve A hectares with N cropping seasons of device life, and let the MRV cost per hectare per season decompose as

$$C = \underbrace{\frac{n_{\text{dev}} p_{\text{dev}}}{AN}}_{\text{hardware}} + \underbrace{c_{\text{conn}}}_{\text{connectivity}} + \underbrace{hw}_{\text{labour}} + \underbrace{\frac{c_{\text{audit}}}{A}}_{\text{verification}}, \quad (23)$$

where n_{dev} and p_{dev} are the number and unit price of devices, h is the labour hours per hectare per season and w the rural wage. Revenue per hectare is

$$R = \kappa \rho, \quad (24)$$

with κ credits issued per hectare and ρ the price per credit. The scheme is economically viable when

$$C \leq \beta R, \quad \beta \in (0, 1), \quad (25)$$

and $\beta = 0.5$ is the level at which monitoring consumes half of the credit revenue. Substituting (24) gives the break-even issuance

$$\kappa_{\min} = \frac{C}{\beta \rho}. \quad (26)$$

The tiers differ in n_{dev} and, more importantly, in which of the three forgeable quantities of (7) they can observe. Tier 0 is the physics layer: $n_{\text{dev}} = 0$ beyond the controller, the log is written on a handset the farmer already owns, and the filters of Sec. 4 run on public weather data at no marginal cost. Tier 1 adds one flow meter at the cooperative pumping station — $n_{\text{dev}} = 1$ for the whole area A rather than one per plot — which observes the total volume. Tier 2 adds one water-level node per cooperative, re-anchoring the twin and restoring the calibration property a measurement-based MRV needs anyway.

None of the three observes timestamps, and by Prop. 1 the first two are exactly invariant under the transformation that pays. The device anchor of Sec. 8 is therefore not an optional fourth tier but a precondition of admissibility: it is produced by the same controller that actuates the pump, so it adds storage and transmission of a record that already exists, and no additional n_{dev} .

Table 4 compares the scheme with the three architectures it replaces. With $\kappa \in [4, 6]$ and $\rho \in [\$15, \$25]$ [9, 10], revenue is $R \in [\$60, \$150]$ per hectare per season, so $\kappa_{\min}$ at $\beta = 0.5$ corresponds to a monitoring budget of \$30–75 per hectare. The full stack sits an order of magnitude below that ceiling, and the incremental cost of the anchor over a controller that is already installed is about 1% of conservative revenue. Table 4 decomposes the comparison against the three architectures it replaces; the connectivity term is shown separately because it is the one that distinguishes a sensor-per-plot design from a single shared gateway.

Remark 2. *The device prices in Table 4 are planning estimates, not quotations. The ordering of the tiers and the break-even inequality (25) do not depend on them; only the absolute margin does. A quotation should replace the estimates before any procurement decision.*

Table 4. MRV cost per hectare per season for the four architectures. Monitoring dominates MRV cost, so the architecture that avoids dedicated sensing wins on labour rather than hardware.

Architecture	Hardware	Connectivity	Labour	Audit	Total \$/ha	Note
Manual observation tube plus photographs	0.5	0.0	8.0	4.0	12.5	current practice under Decision 4801
Water-level sensor per plot	11.3	2.0	1.5	4.0	18.8	accurate, hard to scale to smallholders
Remote sensing plus field verification	0.0	0.0	2.0	6.0	8.0	revisit 2 to 4 days
Attested log, this work	3.6	1.5	0.3	3.0	8.4	gateway timestamp is what makes it admissible

11 Experimental Design

The design is a comparison of four defence layers against nine log types, on a field that deliberately violates every assumption of the twin.

11.1 Removing the circularity

The prior evaluation generated “real” trajectories with the default `SoilProfile`, noiseless ERA5 forcing, irrigation as a one-hour impulse of exactly $\{20, 40\}$ mm, a threshold of exactly 0.80, and a farmer who records the right depth at the right hour. Its 0/60 false accusations were therefore measurements of self-consistency. Here the field is generated by a separate process that differs from the twin along seven axes, each with a published source (Table 5): soil texture drawn from the five FAO-56 Table 2 classes present in the delta; K_c staged per Table 12 rather than fixed at 1.05; irrigation efficiency $\eta \in [0.60, 0.85]$, so the farmer reports water *pumped* while only ηu reaches the root zone; field rainfall drawn as 0.7–1.4 times the grid cell plus Gaussian noise; bund seepage 0–0.02 mm h⁻¹; recording behaviour with 5% forgetting, rounding to 5 mm, a 35% chance of a one-day delay and 15% depth error within $\pm 35\%$; and an initial depletion w_0 drawn from $[0, 0.45]$ rather than assumed to be field capacity. Rounding and delay are applied per *event*, not per hour: applying a 5 mm rounding to a 40 mm event spread over eight hours rounds each 5 mm fragment to zero and silently deletes the record, an artefact that produced empty logs in an early version of this harness.

Weather is hourly ERA5 from the Open-Meteo archive for Can Tho, Soc Trang and Ca Mau over 2024 (water-scarce) and 2025 (water-surplus), no missing values, window 1 March to 29 May (2 160 h), the dry season in which AWD is practised.

11.2 Four defence layers

Layer 1 is the published point-model detector: replay under the default profile, absolute pump-gap threshold $\tau_0 = 5$ mm. Layer 2 is the robust detector of (17) with the proportional threshold of (19), run both on a reduced Θ (12 configurations $\times$ 2 initial states = 24 replays per log, used for

the 10-seed runs) and on the full Θ ($30 \times 5 = 150$ replays, used as a cross-check). Layer 3 is the device anchor of (22). Layer 4, for reference, is physics alone with no volume measurement.

11.3 Nine log types, four of them adversarial

Three are the naive forgeries of the prior work: added events on rainy days, one third of entries inflated to the whole storage span, and a 96 h window of claims deleted. Four are adversarial and, unlike the naive ones, are constructed by an optimiser rather than a fixed recipe: consolidation onto a 24 h and a 96 h grid; a rational adversary that selects whichever consolidation window maximises n_{dry} subject to preserving the total and respecting (13); and the time-shifting hill-climb of Sec. 5, which searches integer offsets in $[-36, 36]$ and accepts a move only if the total is preserved bit-for-bit, every depth stays below the infiltration bound, and n_{dry} strictly increases.

Two harness defects had to be fixed before these numbers meant anything, and both are recorded in the source because they are easy to reintroduce. First, a fraud generator that finds no improving move returns the *original* honest log, so an unconditional count of “attacks caught” silently includes cases where no attack occurred; every adversarial row therefore carries a changed flag and all attack statistics are conditioned on it. Second, comparisons must use the baseline of the variant itself: the paper-condition honest log is generated by a separate call, and scoring it against the field-condition record of a different soil realisation falsely “caught” 56 of 60 honest logs.

11.4 Metrics

For each layer we report the escape rate, i.e. the fraction of forged logs neither tier flags, and the false-accusation rate on honest logs. For credit impact we report n_{dry} of (10), the number of credit-eligible dry phases, against the honest field baseline. A configuration that maximises claimed volume is not automatically profitable: an early adversary that inflated claims by 344 mm drove n_{dry} to zero and would have earned nothing, so volume is reported only as a diagnostic.

12 Results

12.1 The published detector is unsafe, not merely optimistic

Table 6 gives the central negative result. On logs generated under paper conditions the point-model detector accuses nobody, reproducing the prior 0/60. On the same detector applied to field-condition logs it accuses 26.7% of honest farmers at tier 1. The cause is (8) with an absolute threshold: forgetting one entry, rounding, or recording a day late all produce $G > 5$ mm without any wrongdoing, and the measured gap on honest logs reached 75 mm. The proportional threshold of (19) reduces this to 5.0%, and the device anchor to 0/60.

12.2 Physics cannot see the attack that pays

Table 7 is the second negative result, and it is the one that matters economically. Relative to an honest field baseline of 3.23 creditable dry phases per season, time-shifting raises the count to 5.48, an inflation of 69%; under the full uncertainty set the figure is 6.14 against 3.44, or 78%. Consolidation reaches +52% and +53%, and the rational adversary +63% and +70%. Table 8 shows the corresponding escape rates: time-shifting escapes both the point model and the robust physics layer in 91% of cases, and the rational adversary in 100%, exactly as Proposition 2 predicts. The infiltration filter V_5 fires 648 times across the evaluation but exclusively on the crude depth-inflation forgery, never on a time-preserving-volume attack — it is useful, and orthogonal to the threat that pays.

12.3 The anchor closes every class

Table 8 also gives the constructive half. Of 281 audited logs, 116 were genuinely altered by an attack; the device anchor caught 116/116 and accused none of 60 honest field-condition logs. Under the full Θ the figures are 37/37 and 0/18. Naive forgeries are caught by every layer; the distinction is entirely in the adversarial rows, where physics and volume both fail and only the anchor holds.

12.4 The cost of robustness is measurable

Replacing the point model by the intersection over Θ is not free, and the loss must be reported. On omission fraud the point model catches 78.3% of altered logs while the robust rule catches 40% — the same trade-off direction as the proportional threshold, and for the same reason: a defence that refuses to accuse on the strength of one model configuration also refuses some true positives. Naive forgeries remain at 100% under both. We regard this as the correct side of the trade, since a false accusation falls on a farmer with no means of appeal while a missed forgery falls on a verification body that samples anyway, but it is a loss of detection power and not a Pareto improvement.

12.5 Reduced and full uncertainty sets agree

Every conclusion above holds under both the reduced Θ (24 replays per log, 281 logs, 10 seeds) and the full Θ (150 replays, 85 logs, 3 seeds): false accusation 0/60 and 0/18, anchor completeness 116/116 and 37/37, time-shift escape 91% and 100%, credit inflation +69% and +78%. The reduced set was used for the large runs because it contains the extremes of every axis, which is where an innocent world is most likely to exist; the full set confirms that no interior configuration changes a verdict.

Table 5. Experimental configuration. Every value is reproducible from the released code and the public ERA5 forcing.

Component	Value
Weather	Hourly ERA5 via the Open-Meteo archive, no missing values
Stations	3: ca mau, can tho, soc trang
Climate years	2024, 2025 (2024 water-scarce, 2025 water-surplus)
Window	1 March to 29 May (2 160 h), the delta dry season
Soil textures sampled	5 across the FAO-56 Table 2 range
Reality-gap axes	7: soil, K_c stage, η , field rain, seepage, recording behaviour, w_0
Irrigation efficiency η	0.60–0.85 (water reaching roots / water pumped)
K_c regimes	fixed 1.05 (as published) and staged 1.05/1.20/0.90
Recording noise	forget 5%, round to 5 mm, one-day delay 35%, depth error 15% within $\pm 35\%$
Fraud variants	7 (three naive, four adversarial)
Uncertainty set $ \Theta $	$30 = 5 \text{ soils} \times 2 K_c \text{ regimes} \times 3 \text{ efficiencies}$
Initial states $ w_0 $	5, spanning depletion 0 to 0.90
Replays per log	24 (reduced Θ) and 150 (full Θ)
Logs audited	281

Table 6. False accusation on honest logs, the property the scheme is sold on. Paper conditions reproduce the prior 0/60; field conditions expose the absolute gap threshold.

Defence layer	Honest, paper conditions	Honest, field conditions
v1 tier 0 (physics, point model)	0/60	2/60
v1 tier 1 (absolute $G > 5 \text{ mm}$)	0/60	16/60
v2 tier 0 ($\forall$ over Θ)	0/60	0/60
v2 tier 1 ($\tau(n)$ scales with events)	0/60	3/60
Device-anchored command log	0/60	0/60

13 Discussion

13.1 Relation to prior monitoring work

AWD monitoring has been approached from three directions. *Instrumented*: a low-cost wireless water-level sensor for lowland rice under AWD [11], which solves the measurement problem but multiplies device cost per plot. *Remote sensing*: identification of AWD adoption in the Vietnamese Mekong Delta [12], water-status monitoring with ALOS-2 scattering-power decomposition [13], and region-scale machine-learning mapping of water-saving practices [14]; these cover large areas cheaply but revisit every two to four days and cannot resolve the 72 h dry phase that determines credit eligibility. *Programme level*: the one-million-hectare high-quality low-emission rice

Table 7. Creditable dry phases of at least 72 h, the quantity VM0051 and Decision 4801 pay for, against the honest field baseline. The last column counts attacks that actually altered the log.

Log type	<i>n</i>	Creditable dry phases	Inflation	Attacks that changed the log
Honest farmer (field conditions)	60	3.23	reference	—
Added irrigation events	23	4.83	+49%	23/23
Infeasible claimed depth	23	4.04	+25%	23/23
Omission of real commands	23	4.61	+43%	9/23
Claim consolidation, 24 h	23	4.91	+52%	23/23
Claim consolidation, 96 h	23	4.96	+53%	23/23
Time shifting (rational)	23	5.48	+69%	9/23
Rational adversary	23	5.26	+63%	6/23

Table 8. Main result. Escape rate is the fraction of forged logs neither tier flags; the anchor column is conditioned on the log having actually been altered.

Fraud variant	<i>n</i>	Escapes v1	Escapes v2 physics	Caught by anchor
Honest farmer (field conditions)	60	5% accused	0%	0% accused
Added irrigation events	23	0%	0%	23/23
Infeasible claimed depth	23	0%	0%	23/23
Omission of real commands	23	22%	78%	9/9
Claim consolidation, 24 h	23	39%	91%	23/23
Claim consolidation, 96 h	23	39%	91%	23/23
Time shifting (rational)	23	39%	91%	9/9
Rational adversary	23	100%	100%	6/6
All genuinely forged	116	—	—	116/116

programme [15], which had brought over 350 000 hectares under cultivation by 2026 [16], and MRV identified as the missing link for low-emission rice in Southeast Asia [17].

On the agronomic side, meta-analysis quantifies the water and yield effects of AWD [18], while adoption economics [19] and the technical and institutional barriers to scaling [20] explain why instrumented monitoring has not spread. None of these lines evaluates the *reliability of the*

action record itself, which is the question asked here.

The gateway supplying the device-anchored timestamp is the controller of the companion work [6], whose command dispatch runs over a burst-noise channel of the Gilbert–Elliott type [21, 22]. Our own prior scheduling results for safety-critical agricultural IoT networks address the same transmission regime [23, 24], but neither establishes an evidence record, which is the function relied on here. Because that controller already signs every command it issues, the anchor adds storage and transmission of a record that exists regardless — which is what keeps the marginal cost of attestation near zero.

13.2 What survives of the original claim

The premise that motivated this work was attractive: an irrigation command is already a digital event, so MRV evidence should be nearly free. That premise survives, but not in the form we first published. It survives because the controller that issues the command can also sign it, and the signed record is what makes the evidence admissible — not because a hydrological model can detect a lie. Where the original draft claimed that physics attests the log, the correct statement is that physics *constrains* the log: it rules out depths no soil can hold and states no field can reach, which is worth having, and it is silent about the one transformation that pays.

13.3 Why publishing the threshold is not the problem

One might conclude that ϵ should be kept secret. It should not, for two reasons. A verifier must be able to reproduce a decision, which requires published parameters. And secrecy would not help: by Proposition 1 the time-shifting attack preserves the total volume and every individual depth, so it does not need to know ϵ in order to stay beneath the infiltration bound — it needs only to avoid moving water, which is visible to any farmer. The vulnerability is structural, not a consequence of disclosure.

13.4 Relation to remote sensing

Satellite radar detects irrigation events but revisits every two to four days and cannot resolve a 72 h dry phase in mid-period, which is the crediting unit. It therefore cannot substitute for a command record, and it cannot detect time-shifting within a revisit interval either. Its role remains the one VM0051 Appendix 4 assigns it: a per-plot cross-check that lowers the frequency of field verification [4].

13.5 Limitations

1. *Simulated farmers.* Forgeries are injected into simulated reference trajectories. A field pilot is needed to measure behavioural response, in particular whether farmers shift their *actual* pumping once the anchor is known to exist — an attack our model cannot represent, since it changes M rather than L .

2. *The anchor assumes an honest gateway.* Proposition 4 is completeness against log forgery, not against a compromised controller. Device integrity, key custody and replay protection are assumed from the companion work and are not re-established here.
3. *Not applicable to manual irrigation.* Where a sluice is opened by hand there is no command record, so the anchor is unavailable. This excludes a substantial share of delta farms and bounds the addressable population of the scheme.
4. *Θ is a judgement.* Five textures, two K_c regimes, three efficiencies and five initial states bracket the delta but do not exhaust it. A field outside Θ loses the guarantee of Proposition 3, which is exactly the situation the proportional threshold and the query verdict exist to absorb.
5. *Device prices are planning estimates, not quotations.* The tier ordering and the break-even inequality are unaffected; the absolute margin is.
6. *No emission quantification.* The layer produces activity data. Coupling it to a methane model and locally calibrated emission factors is separate work, as is the cost table's assumption that credits scale with n_{dry} .

14 Conclusion

We set out to make rice MRV nearly free by auditing a farmer-written irrigation log against soil-water physics, and the first version of that layer reported no false accusation and complete detection. Both numbers were artefacts of a field generated by the auditing model itself. Rebuilding the field along seven axes of realism, and replacing fixed forgery recipes with an adversary that optimises the credited quantity, changes the conclusion.

The point-model detector falsely accuses 26.7% of honest farmers, because an absolute pump-gap threshold cannot distinguish a forgotten entry from a secret irrigation. A proportional threshold and an intersection over a 150-element uncertainty set reduce that to 5.0% and 0/60 respectively, at a measured cost in detection of omission fraud, from 78.3% to 40%. Meanwhile a volume-preserving, depth-bounded permutation of claimed times inflates creditable dry phases by 69% to 78% while escaping both physical tiers in 91% to 100% of cases, and we prove this is unavoidable for any detector whose verdict is a function of the replayed trajectory: delaying a claim both lengthens the dry phase and increases the deficit available to it, so profit and undetectability share a gradient.

The repair is not hydrological. Matching the claimed log against the gateway's signed, time-stamped command record catches 116/116 genuinely forged logs and accuses none of 60 honest ones, and it costs nothing beyond a controller the companion work already deploys. The honest formulation of the result is therefore narrower and more useful than the one we started with: near-zero-cost MRV for low-emission rice is achievable, but only where a device stands between the farmer and the pump. Three previously published numbers are corrected accordingly — the admissible depth is the range [49.0, 86.0] mm rather than a single 61 mm; the third physical filter cannot activate on dry-season delta rice, clipped depletion peaking at 0.9987 over 150 configura-

tions; and a subthreshold-based absolute gap test is inadmissible. Code, both uncertainty sets, all 281 audited logs and 46 unit tests, one per design hole, are public.

Data, Code and Declarations

Code, the two uncertainty sets, both adversary implementations, and all 281 audited logs are available at <https://github.com/mxuanvan02/TwinGate> under the MIT licence. Weather forcing is public ERA5 reanalysis served by the Open-Meteo archive. This research is funded by Hue University under Grant No. DHH2025-19-07. The authors declare no competing interests. Ethics approval is not applicable: the study uses public reanalysis data and simulation experiments, with no human participants, animals or field interventions.

Tóm tắt và từ khóa (Tiếng Việt)

MRV là nút thắt khiến lượng phát thải giảm được trong lúa phát thải thấp chưa thành tín chỉ mua bán được; riêng giám sát chiếm ba phần tư chi phí đó. Vì mọi hành động quản lý nước đủ điều kiện cấp tín chỉ vốn là một lệnh điều khiển, nhật ký tưới do nông dân tự ghi trông như bằng chứng gần như miễn phí, xếp hạng cao nhất theo Quyết định 4801/QĐ-BNNMT. Chúng tôi dựng lớp chứng thực đó, chứng minh nó không kết tội oan nông dân tuân thủ, rồi tấn công chính nó — phần tấn công là đóng góp của bài. Phát lại nhật ký qua cân bằng nước FAO-56 với ERA5 công khai phơi bày độ sâu tưới mà không đất nào chứa được; nhưng đối thủ chỉ dịch thời điểm lệnh, giữ nguyên tổng lượng và từng độ sâu, đã thổi phồng số pha khô được cấp tín chỉ thêm 69% và vượt cả hai lớp phòng thủ ở 91% trường hợp. Trên 281 nhật ký tại ba trạm Đồng bằng sông Cửu Long, hai năm khí hậu, bộ phát hiện mô hình điểm đã công bố kết tội oan 26,7% nông dân trung thực; còn đối chiếu nhật ký với bản ghi lệnh có chữ ký và dấu thời gian của gateway bắt 116/116 nhật ký giả, không kết tội oan ai. Vật lý không thể đứng một mình; dấu thời gian phải do thiết bị cấp.

Từ khóa: tưới khô ướn luân phiên (AWD), đo đặc báo cáo thẩm định (MRV), tín chỉ các-bon, lúa phát thải thấp, mô hình sinh đôi số.

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Chữ ký tác giả: _____

References

- [1] Balaine N, Carrijo DR, Adviento-Borbe MAA, Linquist BA. Greenhouse Gases from Irrigated Rice Systems under Varying Severity of Alternate-Wetting and Drying Irrigation. *Soil Science Society of America Journal*. 2019;83(5):1533-41.
- [2] Yan X, Yagi K, Akiyama H, Akimoto H. Statistical analysis of the major variables controlling methane emission from rice fields. *Global Change Biology*. 2005;11(7):1131-41.
- [3] Bộ Nông nghiệp và Môi trường. Quyết định 4801/QĐ-BNNMT: Quy trình thí điểm Đo đạc, báo cáo, thẩm định (MRV) trong canh tác lúa chất lượng cao, phát thải thấp vùng Đồng bằng sông Cửu Long; 2025. Hanoi, Viet Nam, 14 November 2025.
- [4] Verra. VM0051 Improved Management in Rice Production Systems, Version 1.1; 2026. Effective 14 July 2026. <https://verra.org>.
- [5] Vanderheyden L, Gayot N, Van den Broeck G. What monitoring, reporting & verification (MRV) systems can reduce costs and enhance scalability of carbon farming? *Journal of Environmental Management*. 2026;401:128885.
- [6] Mai XV, Nguyen TT. A Digital-Twin Safety Gate for Irrigation Command Dispatch over Lossy Networks and Its Relevance to Emission MRV; 2026. Manuscript under review.
- [7] Allen RG, Pereira LS, Raes D, Smith M. Crop Evapotranspiration — Guidelines for Computing Crop Water Requirements. FAO Irrigation and Drainage Paper 56. Rome: Food and Agriculture Organization of the United Nations; 1998.
- [8] Minderhoud PSJ, Erkens G, Pham VH, Bui VT, Erban L, Kooi H, et al. Impacts of 25 years of groundwater extraction on subsidence in the Mekong delta, Vietnam. *Environmental Research Letters*. 2017;12(6):064006.
- [9] Rocky Mountain Institute. Technical Explainer: Carbon Credits for Improved Rice Cultivation; 2025. Issuance range of 4 to 6 credits per hectare under the revised methodology; invalidation of 4.5 million CDM rice credits. Accessed 05/10/2026. <https://rmi.org/technical-explainer-carbon-credits-for-improved-rice-cultivation>.
- [10] Upstream Ag. What Agribusinesses Need to Know About Carbon Credit Pricing; 2026. Rice methane offtake prices of USD 17–18/t (India, Verra) and USD 25/t (Indonesia, JCM), citing Quantum Commodity Intelligence. Accessed 05/10/2026. <https://upstream.ag/p/what-agribusinesses-need-to-know-about-carbon-credit-pricing>.
- [11] Dela Cruz KMS, Ella VB, Suministrado DC, et al. A Low-Cost Wireless Sensor for Real-Time Monitoring of Water Level in Lowland Rice Field under Alternate Wetting and Drying. *Water*. 2022;14(24):4128.
- [12] Lovell RJ. Identifying Alternative Wetting and Drying (AWD) Adoption in the Vietnamese Mekong River Delta. *ISPRS International Journal of Geo-Information*. 2019;8(7):312.

- [13] Islam MR, Oyoshi K, Takeuchi W. Alternate Wetting and Drying Irrigated Rice Paddy Field Water Status Monitoring with ALOS-2 Three-Component Scattering Power Decomposition. *Remote Sensing*. 2026;18(8):1183.
- [14] Shah A, Singh R, Zaytar A, Tadesse GA, Robinson C, Tafti N, et al. Machine Learning for Sustainable Rice Production: Region-Scale Monitoring of Water-Saving Practices in Punjab, India. *arXiv preprint arXiv:250708605*. 2025.
- [15] Thủ tướng Chính phủ. Quyết định 1490/QĐ-TTg: Phê duyệt Đề án “Phát triển bền vững một triệu héc-ta chuyên canh lúa chất lượng cao và phát thải thấp gắn với tăng trưởng xanh vùng đồng bằng sông Cửu Long đến năm 2030”; 2023. Hanoi, Viet Nam, 27 November 2023.
- [16] Ministry of Agriculture and Environment of Viet Nam. More than 350,000 hectares of low-emission rice cultivated in Viet Nam; 2026. Programme progress, 11 MRV pilots in 2025, emission reduction of 3–4 t CO₂e per hectare, farmer income up 13.4%. Accessed 05/10/2026. <https://en.mae.gov.vn/more-than-350000-hectares-of-lowemission-rice-cultivated-in-viet-nam-9192.htm>.
- [17] Gonzalez T, Concepcion G. Why MRV is the missing link for low-emission rice in Southeast Asia; 2026. *Rice Today*, IRRI, 16 April 2026. ricetoday.irri.org.
- [18] Carrijo DR, Lundy ME, Linquist BA. Rice yields and water use under alternate wetting and drying irrigation: A meta-analysis. *Field Crops Research*. 2017;203:173-80.
- [19] Lampayan RM, Rejesus RM, Singleton GR, Bouman BAM. Adoption and economics of alternate wetting and drying water management for irrigated lowland rice. *Field Crops Research*. 2015;170:95-108.
- [20] Enriquez Y, Yadav S, Evangelista G, et al. Disentangling Challenges to Scaling Alternate Wetting and Drying Technology for Rice Cultivation. *Frontiers in Sustainable Food Systems*. 2021;5.
- [21] Gilbert EN. Capacity of a Burst-Noise Channel. *Bell System Technical Journal*. 1960;39.
- [22] Elliott EO. Estimates of Error Rates for Codes on Burst-Noise Channels. *Bell System Technical Journal*. 1963;42.
- [23] Mai XV, Le DMP, Dang TN, Truong KD, Duong DG, Nguyen TT. Self-Tuning Risk-Adaptive Bandwidth Scaling for Safety-Critical Smart-Agriculture IoT Networks; 2026. Manuscript under review.
- [24] Mai XV, Le DMP, Dang TN, Truong KD, Nguyen HS, Nguyen TT. Threshold-Aware Probe-Then-Transmit Scheduling for Safety-Critical IoT Monitoring; 2026. Manuscript under review.